ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

Choose $x_1 = \theta$ and $x_2 = \dot{\theta}$. Then $\dot{x}_1 = x_2$ and $\dot{x}_2 = [\tau - b\, x_2 - mg\ell \sin(x_1)]/J$. At equilibrium:

$$x_2 = 0 and_0 = mg\ell sin\theta_e$$

Since |sin θ_e| less than or equal to 1, an equilibrium exists when $|\tau_0|$ less than or equal to $mg\ell = 2.4525$ N m.

At rest $\dot{\theta} = 0$ and $\ddot{\theta} = 0$, so the damping term vanishes. Rearranging gives $\tau_0 = mg\ell \sin\theta_e$. Here $mg\ell = 1 \times 9.81 \times 0.25 = 2.4525$ N m. Since $-1 \leq \sin\theta_e \leq 1$, the required torque must lie between -2.4525 and $+2.4525$ N m.

Q2(b) [2 marks]

Q2(c) [4 marks]

At rest, 50(π/2-θss) = 2.4525. This gives θss = 1.5218 rad = 87.19°, so error approximately 2.81°.

Q2(d) [2 marks]

With gravity compensation, the gravity term cancels the plant's $mg\ell \sin\theta$ term. At rest, Kp(θd-θss)=0, so θss=θd and the steady-state error is zero. The linearised gravity stiffness term is also cancelled, so the closed-loop poles generally change; the stiffness contribution $mg\ell \cos\theta_d$ is removed.

The compensation term cancels the nominal gravity torque in the plant model. At steady state the proportional term must then be zero, which requires $\theta_d - \theta_{ss} = 0$. This does not mean the transient response is unchanged: removing the gravity derivative changes the linearized stiffness.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

The Tustin coefficients are y[k]=0.881765y[k-1]+0.0591175(x[k]+x[k-1]). The DC gain is 0.0591175/(1-0.881765)=0.5.

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. I obtain p(s)=s^2+(ℓ_1+2)s+(ℓ_1+ℓ_2). Matching $(s+10)^2$ gives ℓ_1=18 and ℓ_2=82.

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

Poles are -2±2j and Nff=8.

Q4(b) [2 marks]

The nominal controller has unity DC gain, so θss=1 and the tracking error is zero even after the actuator gain changes.

Q4(c) [4 marks]

I obtain s^3+(a+bk_2)s^2+(c+bk_1)s-bK_I. Desired polynomial is s^3+8s^2+24s+32, so k_2=6, k_1=18, K_I=-32.

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.